

University of the Witwatersrand
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Literature Review

Safe AI via Evolutionary Algorithms

Honours Research Project

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Contents

1	Introduction	2
2	Background	2
2.1	Safe Reinforcement Learning: Formalisms and Benchmarks	3
2.2	Evolution Strategies as a Scalable Alternative to Gradient-Based RL	4
2.3	Quality Diversity: From Single Optima to Behavioural Archives	5
2.4	Parameter-Efficient Fine-Tuning	6
2.5	Policy-Gradient Methods for Alignment	7
3	Related Works	8
3.1	Evolutionary Methods for Large Language Model Alignment	8
3.2	Adversarial Attacks on Aligned LLMs	9
3.3	Synthesis and Research Gap	10
4	Conclusion	11
	References	11

1 Introduction

Modern large language models (LLMs) are aligned to human preferences through reinforcement learning from human feedback (RLHF). RLHF first trains a small *reward model* that scores candidate responses against human preference data, then uses a gradient-based RL algorithm (typically Proximal Policy Optimization [9]) to fine-tune the LLM so that its outputs maximise this reward. The reward model thus stands in for human judgement: the LLM never sees humans directly, only the proxy.

The trouble is that any proxy can be gamed. Aligned LLMs routinely exhibit *reward hacking*: they discover outputs that score highly on the reward model without genuinely being safe or helpful. A model trained to be “concise” learns to emit near-empty responses; a model trained to be “helpful” learns flattery. Reward hacking is the central safety challenge in deployed AI today.

Reward hacking is not specific to LLMs. It is the LLM instance of the broader *safe reinforcement learning* (safe RL) problem: an agent optimising a scalar return must nevertheless avoid a separately specified notion of unsafe behaviour. In safe RL, “unsafe” typically means violating a hand-coded constraint (a robot tipping over, a vehicle leaving its lane). In LLM alignment, “unsafe” means whatever the reward model fails to penalise. The shape of the problem is the same.

Two adjacent literatures bear directly on this problem. The first is safe RL itself, which has converged on Constrained Markov Decision Processes (CMDPs) as the dominant formalism, supported by benchmarks and algorithm libraries that are now standard reference points. The second is *evolutionary strategies* (ES): a family of population-based, gradient-free optimisers. Rather than following a gradient, ES explores parameter space by sampling many slightly perturbed copies of a policy and moving towards copies that score higher. ES has recently been scaled from small control problems to the full parameter space of billion-parameter LLMs.

These two literatures developed largely independently, but a small, very recent body of work has begun to connect them. Quality–diversity methods borrow ES gradient estimators to populate behavioural archives, and ES is being used as a gradient-free alternative to RLHF for LLM alignment. This review surveys that intersection.

Section 2 introduces the foundational threads: safe-RL formalisms (Section 2.1), evolution strategies (Section 2.2), quality–diversity methods (Section 2.3), parameter-efficient fine-tuning via LoRA (Section 2.4), and policy-gradient methods (Section 2.5). Section 3 turns to related work most directly bearing on the CoEvoRL project: recent papers applying ES to LLM alignment at scale (Section 3.1), gradient-based adversarial attacks on aligned LLMs (Section 3.2), and a synthesis identifying the gap CoEvoRL aims to fill (Section 3.3).

2 Background

This section surveys the foundational threads on which the rest of the review builds: safe-RL formalisms and benchmarks, evolution strategies as scalable gradient-free RL alternatives, quality–diversity methods that maintain archives of behaviourally distinct solutions, parameter-efficient fine-tuning of large pre-trained models, and the policy-gradient methods that underpin contemporary RLHF. Each thread is treated independently here; their composition with recent ES-based LLM alignment work is taken up in Section 3.

2.1 Safe Reinforcement Learning: Formalisms and Benchmarks

Safe RL studies how to train policies that maximise a return signal while also satisfying additional safety requirements. The dominant formalism is the *Constrained Markov Decision Process* (CMDP), which augments a standard MDP with one or more *cost* functions. The agent then seeks a policy that maximises expected return subject to expected cost remaining below a user-specified threshold. Concretely, “return” might be the reward for reaching a goal, while “cost” counts collisions or constraint violations.

Ji et al.’s *Safety-Gymnasium* [1] is the most complete instantiation of this framework in continuous control. It packages over 54 task configurations spanning single-agent, multi-agent, and vision-only settings, with simulated robots (Point, Car, Doggo, Racecar, Ant) navigating environments under velocity, position, and contact constraints. Tasks are formalised as CMDPs (or, in multi-agent settings, Constrained Markov Games), and the suite replaces the discontinued `mujooco-py` back-end of the earlier Safety Gym with native MuJoCo support.

Alongside the benchmark, the authors release *SafePO*, a single-file library of sixteen state-of-the-art safe-RL algorithms covering Lagrangian, projection-based, first-order, and second-order methods. The headline empirical finding is negative: no single algorithm dominates across the benchmark. Lagrangian methods produce strong numbers but are notoriously hyperparameter-sensitive; second-order methods excel at navigation but underperform on velocity tasks. The authors flag two acknowledged limitations: sim-to-real transfer is unaddressed, and human-factor concerns in collaborative settings are not modelled.

A third absence is more consequential for this review: every algorithm in SafePO is gradient-based. Derivative-free approaches, including ES and quality-diversity algorithms, are entirely absent, leaving open the question of whether population-based methods can compete on these constraint-satisfaction tasks.

A different response to the complexity of the CMDP framework is provided by Nangue Tasse et al. [4]. ROSARL argues that requiring users to specify cost functions and thresholds is itself a limitation, and proposes that safety can be encoded entirely through the reward signal. The central theoretical contribution is the *Minmax penalty*: the smallest penalty applied at unsafe states that is provably sufficient to guarantee safe optimal policies, regardless of the task reward.

The size of this penalty depends on two structural properties of the environment. The *diameter* is the longest expected time to reach an absorbing state, and the *controllability* is the minimum probability of reaching a safe goal state. Computing the exact Minmax penalty is NP-hard, so the authors derive a model-free approximation that estimates the penalty online from value-function bounds and can be dropped into any standard RL pipeline.

Empirically, reward-only agents trained with this penalty produce safer behaviour than several CMDP baselines (CPO, TRPO-Lagrangian, Saute-TRPO) in Safety Gym environments. The advantage is largest under high environmental stochasticity, where the CMDP baselines degrade into risky short-trajectory policies.

ROSARL’s theoretical scope is deliberately narrow: it targets unsafe *terminal* states in undiscounted stochastic shortest-path MDPs, and the continuous-control experiments therefore violate the strict conditions of the theorem. Even so, the framing matters for what follows. Reward-only safe RL operates entirely on scalar fitness, which is precisely the signature evolutionary optimisers consume. The Minmax penalty could therefore be plugged into ES with no change to the optimiser, while CMDP-style constraints would require ES to handle two scalar

signals (reward and cost) and a threshold. Both framings reappear in Section 3.3; ROSARL’s reward-only view composes most naturally with the evolutionary methods covered next.

2.2 Evolution Strategies as a Scalable Alternative to Gradient-Based RL

Salimans et al.’s *Evolution Strategies as a Scalable Alternative to Reinforcement Learning* [5] is the canonical reference for modern ES-based policy optimisation. Their algorithm, a variant of Natural Evolution Strategies now called OpenAI-ES, works as follows: maintain a centroid parameter vector for the policy network; each iteration, perturb it with isotropic Gaussian noise to produce a population of perturbed copies; evaluate each copy in the environment; update the centroid in the direction of the higher-scoring perturbations. There is no backpropagation through the policy or the environment, and the only signal exchanged between workers is the scalar episodic return.

The first major contribution is a communication strategy that makes ES embarrassingly parallel. Workers agree on a random seed schedule in advance, regenerate the perturbations locally from the seeds rather than receiving them over the wire, and exchange only scalar returns. Bandwidth is therefore independent of network size, enabling near-linear scaling to more than 1,440 parallel CPU workers, roughly an order of magnitude beyond what gradient-based RL can sustain.

The second contribution is empirical parity with mainstream RL. OpenAI-ES matches A3C on 23 of 51 Atari games at comparable total compute, solves the MuJoCo 3D humanoid task in under ten minutes of wall-clock time, and discovers qualitatively distinct gaits in locomotion tasks that gradient-based methods do not find. The exploration induced by parameter-space perturbation is therefore not a substitute for action-space noise: it appears to be a genuinely different kind of search.

The third, and most theoretically interesting, is a variance analysis. The variance of the ES gradient estimate is independent of the episode length T , whereas the variance of the standard policy-gradient estimator grows linearly in T . ES therefore becomes *relatively* more favourable as horizons lengthen, and is naturally robust to action frequency, sparse rewards, and reward delays, all consequences of treating the episode as a single black box.

The authors are candid about the costs. ES uses three to ten times more environment frames than A3C to reach comparable performance, and its wall-clock competitiveness rests on massive parallelism rather than sample efficiency. Without virtual batch normalisation, perturbed networks frequently degenerate into constant policies. More importantly for what follows, OpenAI-ES uses a fixed isotropic Gaussian perturbation distribution that cannot adapt to the local geometry of the loss surface, and converges on a single optimised policy. Both choices limit behavioural diversity, a limitation that the next subsection’s quality–diversity methods address directly and that the adaptive covariance of CMA-ES partially addresses.

Where OpenAI-ES is engineered for parallelism, CMA-ES [8] is engineered for sample efficiency. CMA-ES samples each candidate from a multivariate Gaussian $\mathcal{N}(m, \sigma^2 C)$, where m is the population mean, σ is a global step size, and C is a full covariance matrix that captures correlations between parameters. Each generation, m moves towards the weighted mean of the top- μ candidates, while C is refined to align with directions in which good solutions are found, so future samples concentrate on promising regions of parameter space.

Step size is controlled by a separate mechanism that compares the length of accumulated suc-

cessful steps against what would be expected under random selection: longer runs of correlated progress inflate σ , while shorter ones shrink it. The result is a black-box optimiser whose only hyperparameters (recombination weights, learning rates, dampings) are functions of the search dimension and population size, giving CMA-ES its widely-noted near-tuning-free reputation.

Two further structural properties make CMA-ES the workhorse of derivative-free continuous optimisation. The first is *invariance*: under translation, rotation, or any order-preserving transformation of the objective, CMA-ES’s trajectory is unchanged. The second is *implicit geometry*: by tracking C , CMA-ES estimates an inverse-Hessian-like preconditioner from rank information alone, without ever differentiating the objective.

The catch is dimensionality. The per-generation eigendecomposition of C costs $O(n^3)$ in the search dimension n , effectively capping full-covariance CMA-ES at a few thousand parameters. This dimensionality wall is precisely why both ESSA (Section 3.1) and the CoEvoRL attacker apply CMA-ES not to raw model weights but to a compressed projection: ESSA via SVD on LoRA matrices, CoEvoRL via a Johnson–Lindenstrauss random projection from a low-dimensional CMA search space onto the much larger LoRA delta. Restricted-CMA variants (separable CMA, sep-CMA, VxD-CMA) approximate C with diagonal or low-rank structure for higher dimensions, but trade away the algorithm’s full adaptive geometry.

2.3 Quality Diversity: From Single Optima to Behavioural Archives

Where OpenAI-ES collapses an entire population onto a single centroid, *quality–diversity* (QD) algorithms keep the population explicit. The aim shifts from finding one best policy to maintaining many qualitatively different policies, each the best in its own behavioural niche.

The canonical QD algorithm is MAP-Elites, introduced by Mouret and Clune [3]. It maintains a discretised grid over a user-chosen *feature space* (for example, the proportion of time a walking robot spends on each leg). Each grid cell stores the highest-fitness solution discovered so far whose features fall in that cell. New candidates are produced by mutating existing cell elites; an offspring is placed in the cell its features map to, replacing the incumbent only if it scores higher.

The output is therefore not a single optimum but a structured archive showing how performance varies across the chosen behavioural dimensions. Mouret and Clune’s key empirical observation, replicated across three domains (neural-network image generation, simulated soft-robot morphologies, and a real soft robotic arm), is what they call the *stepping-stone effect*: top-performing cells are typically reached via long lineages that pass through cells far from the final region of interest. Diversity is not just a presentation device but an optimisation device, because behaviourally distinct intermediate solutions are essential for finding the eventual best one.

MAP-Elites has two well-known limitations. The feature grid suffers from the curse of dimensionality (cell count grows exponentially with the number of feature dimensions), and feature descriptors must be hand-designed in advance. A more consequential issue is that MAP-Elites is gradient-free in a strong sense: offspring come from random mutation, even where useful gradient information is available.

Tjanaka et al.’s *CMA-MEGA* [2] closes this gap in the RL setting. CMA-MEGA retains the MAP-Elites archive but generates candidates by sampling in a space of *gradient coefficients* (linear combinations of estimated objective and measure gradients), with CMA-ES driving

the search. Because true gradients are unavailable in RL, both must be estimated. CMA-MEGA (ES) estimates both via the OpenAI-ES gradient estimator, making the algorithm fully model-free; CMA-MEGA (TD3, ES) uses the actor-critic method TD3 for the objective gradient while keeping ES for the measure gradient.

On four PyBullet locomotion tasks, CMA-MEGA (TD3, ES) matches the state-of-the-art PGA-MAP-Elites, while the fully ES-based variant underperforms. The diagnosis is illuminating: in reward-shaped domains, the ES-only variant treats the objective as a pure black box and gets stuck in local optima (for example, agents that learn to stand still to maximise a “stay upright” bonus), missing the MDP structure that TD3’s critic exploits.

The cautionary implication is that pure ES is not uniformly dominant; where gradient information is cheap, hybrid approaches do better. The constructive implication, which the QD literature has not yet pursued at scale, is that MAP-Elites’ feature space need not be confined to locomotion descriptors. Any measurable attribute of a policy can serve as a feature dimension, including ones directly relevant to safety: constraint-violation rate, detectability by a classifier, or distance from a nominal “safe region” of behaviour. A safety-dimensional MAP-Elites grid would illuminate the safety–performance trade-off in the same way the original illuminates the morphology–performance trade-off, a connection that links QD directly to the safe-RL literature of Section 2.1 and that remains largely unexploited.

2.4 Parameter-Efficient Fine-Tuning

Full fine-tuning of frontier large language models is prohibitive in storage, memory, and serving cost. A task-specific copy of GPT-3 holds 175 billion trainable parameters, and the Adam optimiser state inflates the memory budget several-fold beyond that. *Low-Rank Adaptation* (LoRA) of Hu et al. [10] is the parameter-efficient fine-tuning method that has come to dominate the field.

The idea is simple. For each pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$ that you want to adapt, freeze it and learn instead a small additive update $\Delta W = BA$, where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ have rank $r \ll \min(d, k)$. B is initialised to zero and A to a Gaussian, so training begins from exactly the pre-trained model. The forward pass becomes $h = W_0x + (\alpha/r)BAx$ for a fixed scaling α .

At inference time, $W_0 + BA$ can be merged back into a single matrix, recovering the original architecture’s FLOPs and latency exactly. This property distinguishes LoRA from earlier adapter methods that injected new residual blocks into the network depth and therefore paid a permanent inference cost.

The empirical claims are striking. On GPT-3 175B, LoRA at rank $r = 4$ applied only to attention query and value projections matches or exceeds full fine-tuning while training fewer than five million parameters, a roughly 37 000-fold reduction. Comparable parity holds across RoBERTa, DeBERTa, and GPT-2 on the GLUE benchmark. The accompanying analysis shows that the effective rank of ΔW is very low, often $r = 1$ or $r = 2$, even when the underlying weight dimension exceeds twelve thousand.

This rank-deficiency finding is the bridge between LoRA and the evolutionary methods of this review. If fine-tuning genuinely operates in a low-dimensional subspace of full parameter space (the *intrinsic-dimensionality* hypothesis on which LoRA’s design rests), then evolution

strategies are not facing a billion-parameter problem at all. They are facing a problem of perhaps a few million parameters (the LoRA delta) embedded in a billion-parameter model.

ESSA exploits this directly, running CMA-ES on SVD-compressed LoRA matrices; the CoEvoRL attacker exploits it by treating the LoRA parameters as the search space and applying a further random-projection compression on top. Both designs depend on LoRA both as a tractable parameterisation of the search space and as the inference-merging mechanism that allows population-based evaluation without per-individual architectural overhead.

2.5 Policy-Gradient Methods for Alignment

The dominant gradient-based alternative to evolutionary search is policy-gradient reinforcement learning, and the canonical algorithm in modern practice is *Proximal Policy Optimization* (PPO) of Schulman et al. [9]. PPO addresses two persistent failure modes of vanilla policy gradient: the data inefficiency of single-pass updates, and the destructive variance of unrestricted reuse of a sampled trajectory across multiple training epochs.

The core idea is to define an importance ratio $r_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$, which measures how much the new policy’s probability of an action differs from the old policy’s. PPO then optimises a *clipped* surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t \hat{A}_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right],$$

with a small clip threshold (typically $\epsilon = 0.2$). The clip removes the gradient incentive to push r_t outside the trust region, and the outer minimum makes the objective a pessimistic lower bound. Together, they permit many epochs of minibatch SGD on a single rollout without destructive policy updates.

Compared to TRPO, PPO retains the empirical monotonicity of trust-region methods using only first-order optimisation: no constraint solver, no Hessian-vector products, just standard automatic differentiation. Empirically, PPO matches or exceeds TRPO on MuJoCo continuous control and dominates A2C on Atari sample efficiency, with a single shared set of hyperparameters across tasks.

PPO matters to this review because it is the underlying optimiser of every contemporary alignment pipeline against which evolutionary methods are compared. Modern RLHF replaces the environment reward with a preference-based reward model and adds an adaptive KL penalty against a reference policy, but the underlying update is PPO. Recent variants such as GRPO and DPO are descendants of the same lineage.

When Qiu et al. [7] report that ES Pareto-dominates GRPO on the conciseness reward–KL trade-off (Section 3.1), they are implicitly reporting that distribution-based search Pareto-dominates a PPO-class point estimate. The CoEvoRL design proposes PPO as the inner-loop optimiser within an outer-loop CMA-ES: each individual in the CMA-ES population is fine-tuned for a small number of PPO updates against the current monitor before its scalar fitness is reported back to the outer loop. This nests the algorithmic contributions of Sections 2.2 and 2.5 in a manner that, to the best of this review’s knowledge, no surveyed work has attempted.

3 Related Works

The foundational threads surveyed in Section 2 have, until very recently, developed in isolation from the alignment of large language models. This section turns to the related work that does begin to connect them: two recent papers that apply ES at LLM scale (Section 3.1); the canonical gradient-based adversarial attacks on aligned LLMs (Section 3.2); and a synthesis that locates the CoEvoRL research agenda within the gap that remains between these threads (Section 3.3).

3.1 Evolutionary Methods for Large Language Model Alignment

Applying ES to LLM alignment seems, at first, implausible. OpenAI-ES was demonstrated on networks with a few hundred thousand parameters; billion-parameter LLMs are five orders of magnitude larger, and the folklore view is that ES-style perturbation becomes useless in such high-dimensional parameter spaces. Two recent papers (Korotyshova et al. [6] and Qiu et al. [7]) overturn this view, via nearly opposite design choices, and together they bracket the emerging design space for evolutionary LLM fine-tuning.

ESSA [6] takes the compression route. Fine-tuning is restricted to Low-Rank Adapters (LoRA) [10], and the LoRA matrices are further decomposed via singular value decomposition (SVD); CMA-ES then optimises only the singular values of each adapter matrix. The search space is therefore drastically smaller than the full parameter space, which keeps CMA-ES tractable: its covariance matrix would otherwise scale quadratically with dimensionality.

ESSA inherits OpenAI-ES’s shared-seed communication, so only seeds and scalar rewards traverse the network, and operates entirely in quantised inference mode (BFLOAT16 / INT8 / INT4) with minimal accuracy loss. On Qwen2.5-32B fine-tuned against PRM800K, ESSA reaches near-optimal accuracy twice as fast as GRPO on 16 GPUs and six times as fast on 128 GPUs, while remaining stable across LoRA ranks 2–16 and population sizes 24–96.

Qiu et al. [7] take the opposite route. Rather than compress the search space, they search the full parameter space of LLMs up to 8B parameters with a deliberately vanilla NES variant: fixed isotropic Gaussian noise, no rank transformation, no mirrored sampling, no Adam. The remarkable empirical finding is that a population of only $N=30$, roughly 300 times smaller than OpenAI-ES’s populations on much smaller networks, suffices to optimise billion-parameter models. The authors attribute this to the low intrinsic dimensionality of LLM fine-tuning.

Empirically, the results are competitive or better than gradient-based RL. On the Countdown symbolic-reasoning task, ES substantially outperforms the best RL baselines (60.5% versus 43.8% on Qwen2.5-3B; 66.8% versus 57.5% on Qwen2.5-7B). On mathematical reasoning it is competitive with state-of-the-art RL methods (33.1% versus 30.5% on Minerva Math; 78.6% versus 78.0% on MATH500). Implementation-wise, the paper stores random seeds rather than noise tensors, perturbs and restores parameters layer by layer, and applies decomposed updates, keeping peak memory only marginally larger than standard inference.

Two findings from the Qiu et al. paper matter more for this review than the headline numbers. First, ES *naturally* avoids reward hacking. In conciseness fine-tuning, GRPO without a carefully tuned KL-divergence penalty collapses into producing near-empty outputs that game the reward; ES, with no KL penalty at all, produces a Pareto front that dominates GRPO’s: higher reward at lower KL divergence.

The authors frame this as a structural property of optimising a *distribution* over policies rather than a single point estimate. Reward-hacking solutions tend to be isolated spikes in the reward landscape, and a distribution-based optimiser is blind to them: a sample-efficient exploit that pays off at one point does not translate into expected-return gains across the perturbation neighbourhood.

Second, ES is substantially more consistent than RL across both runs and base models. The same fixed hyperparameters work across all tested models, while each RL method requires per-model tuning.

What both papers share is an *asymmetric* framing of alignment: a single LLM is optimised, via ES, against a fixed reward signal. ESSA targets a mathematical-reasoning reward; Qiu et al. target symbolic-reasoning, mathematical-reasoning, or conciseness rewards. Neither paper explores adversarial settings in which the reward is itself produced by another optimising agent, and neither imposes CMDP-style constraints on the evolutionary search. The safety-relevant properties they identify (emergent resistance to reward hacking and robustness across base models) are therefore documented only in the static, single-sided regime.

3.2 Adversarial Attacks on Aligned LLMs

Aligned LLMs (ChatGPT, Claude, Llama-2-Chat, and their open-source counterparts) are explicitly fine-tuned to refuse harmful queries, but the alignment is leaky in the adversarial sense: small perturbations to a query reliably elicit responses the model would otherwise refuse.

The canonical automated attack is the *Greedy Coordinate Gradient* (GCG) method of Zou et al. [11]. GCG searches for a discrete-token adversarial suffix that, when appended to any harmful query, maximises the probability that the target LLM produces an affirmative prefix (“Sure, here is. . .”). The intuition is that if the model can be coerced into starting with “Sure”, it will continue along the harmful path rather than refuse.

The algorithm operates in two interleaved steps. At each iteration, it computes the gradient of the loss with respect to one-hot token indicators at every suffix position, and takes the top- k candidate replacement tokens per position by negative gradient. It then samples a batch of (position, candidate) pairs, evaluates each, and commits the loss-reducing replacement.

Optimised against a small set of harmful prompts on white-box source models (Vicuna-7B/13B and Guanaco-7B), the resulting suffix achieves an 88% target-elicitation attack success rate on Vicuna, against 25% for the prior AutoPrompt method. More strikingly, the suffix transfers non-trivially to closed-source production systems including ChatGPT, Bard, and Claude, despite the attacker having no gradient access to those models. The companion AdvBench benchmark and Attack Success Rate (ASR) metric have since become the standard evaluation harness for jailbreak attacks.

GCG’s design is illuminating for this review precisely because it is the high-water mark of a particular paradigm. The optimiser is gradient-based on white-box source models. The search space is discrete tokens. The objective is a single fixed reward (the affirmative-prefix probability). The output is a single suffix that compresses many harmful behaviours onto one narrow region of the attack-prompt manifold.

Subsequent work in the same paradigm relaxes individual constraints but preserves the overall asymmetry. PAIR adds an LLM-judge feedback loop to operate against black-box targets; AutoDAN uses a genetic algorithm to mutate full natural-language prompts rather than discrete

suffixes; their descendants extend further. None of them maintains a population of attackers under selection pressure from a co-evolving defender, and none imposes constraint-satisfaction structure on the attack search.

The consequence is that the failure modes surfaced by gradient-based red-teaming form a comparatively small and structurally homogeneous set. This both makes them easier to patch in subsequent fine-tuning rounds and limits their value as a training signal for genuinely robust safety monitors. This observation is the empirical anchor for the gap argument developed in Section 3.3.

3.3 Synthesis and Research Gap

Taken together, the surveyed literature supplies three self-contained components that bear on the problem of producing safe, adversarially robust AI via evolutionary optimisation:

1. **A formalism for safety.** Safety-Gymnasium [1] provides the CMDP formalism and a standardised benchmarking infrastructure; ROSARL [4] provides a reward-only alternative in which safety is encoded through a provably sufficient Minmax penalty, and derives that penalty’s dependence on environment diameter and controllability. Both framings reduce safety to signals (scalar returns and/or scalar costs) that black-box optimisers can consume without modification.
2. **Gradient-free, population-based search.** OpenAI-ES [5] establishes that evolutionary search can match gradient-based RL on non-trivial control and game-playing tasks, and scales near-linearly with compute. MAP-Elites [3] and its gradient-informed successor CMA-MEGA [2] add a structural commitment to *behavioural diversity*: both algorithms preserve archives of qualitatively distinct solutions, and CMA-MEGA demonstrates how ES gradient estimators can be grafted onto such archives in RL settings.
3. **Evolutionary search at LLM scale.** ESSA [6] and Qiu et al. [7] establish, via opposite design choices (aggressive LoRA+SVD compression versus full-parameter search with small populations), that ES is a viable post-training method for billion-parameter LLMs. Qiu et al. additionally document that ES appears to resist reward hacking as an emergent property of distribution-based optimisation, and produces more consistent results across runs and base models than RL.

What the surveyed literature does *not* contain is any combination of these three components. Three adjacent gaps are worth naming.

First, as surveyed in Section 3.2, existing LLM red-teaming methods are gradient-based, single-sided, and converge to narrow regions of the attack-prompt space. An evolutionary red-teamer that maintains a diverse population of attack strategies has no analogue in the published literature.

Second, the evolutionary LLM fine-tuning work of Section 3.1 is one-sided: a single LLM is optimised against a fixed external reward, with no adversarial pressure and no explicit safety constraints.

Third, the safe-RL machinery of Section 2.1 has not been transposed to LLM fine-tuning at all; CMDPs and ROSARL-style penalties remain studied almost exclusively in robotic control.

These three gaps intersect at a specific design: a *co-evolutionary* system in which a population of evolutionary attacker LLMs and a population of safety-monitor classifiers are optimised

simultaneously, under a CMDP constraint that penalises detection probability rather than expected cost, with the constraint threshold itself co-evolved via a ROSARL-style mechanism. This is the design at the heart of the CoEvoRL project.

Every individual ingredient (CMA-ES [8], LoRA [10], CMDP, MAP-Elites-style archival, self-adapting penalties) is established in the literature surveyed above. What is novel, and what this project proposes, is their composition in an adversarial LLM setting.

4 Conclusion

The literature surveyed in this review spans three threads: safe-RL formalisms (CMDPs, reward-only alternatives), evolutionary strategies and their quality-diversity extensions, and the very recent application of ES to LLM alignment. Each thread is internally mature; each has established methods, benchmarks, and reproducible empirical claims.

What has not been established is the combination. Composing these ingredients into a co-evolutionary, constrained, evolutionary LLM fine-tuning system would aim at two outcomes: safety monitors that generalise to attacks they were never trained against, and the surfacing of failure modes that gradient-based red-teaming structurally cannot reach.

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Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ☒
2. Sourcing Related Work (summarising, identifying sources) ☒
3. Methods and Experiment Design (experiment setup, model tuning) ☒
4. Data Analysis (presentation, coding, interpretation) ☒
5. Theoretical Development (theorem proving, conceptual analysis) ☒
6. Code Development (generating algorithms, writing scripts) ☒
7. Presentation (rendering graphics, formatting) ☒
8. Editing (grammar, readability) ☒
9. Writing (text generation, document structuring) ☒
10. Citation Formatting (structuring, organising) ☒

If other uses were involved, please specify below:

Generative AI tool used (list all)	Used for?

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 2699783

Candidate signature: 

Date: 29 March 2026